

Sheet 2 - calculus

Q10 graded

1. Suppose $p^2/q^2 = 2$, where $a^2 := p^2/q^2$ is a fraction in its simplest form.

Then $p^2 = 2q^2$ for some integers p, q . So q^2 is an int., and thus p^2 must be even. Any number whose square is even is itself even. So p is even; let $p = 2m$ for an int. m .

Then $q^2 = p^2/2 = 2m^2$ is even; by the same argument q is even.

But then a was not in its simplest form. $\times$
 $\therefore \sqrt{2}$ is not a rational number.

Now consider the set $S := \{x \in \mathbb{Q} : x^2 < 2\}$ or $x < 0$ and $T := \{x \in \mathbb{Q} : x^2 > 2\}$ and $x > 0$.

There does not exist any $y \in \mathbb{Q}$ s.t.

$\forall s \in S \forall t \in T \quad s \leq y \leq t$. for So $\mathbb{Q}$ is not complete,

since the only such y would be $\sqrt{2} \notin \mathbb{Q}$.

2. $\|x+y\| := \sqrt{\sum_{i=1}^n (x_i+y_i)^2}$

$$\therefore \|x+y\|^2 = \sum_{i=1}^n 2x_i y_i + x_i^2 + y_i^2$$

$$= \|x\|^2 + \|y\|^2 + 2 \sum x_i y_i$$

$$\text{and } (\|x\| + \|y\|)^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|.$$

Norms are non-negative, so $\|x\|\|y\| \geq 0$. Since

$|2 \sum x_i y_i| \leq \|x\|\|y\|$ by Cauchy-Schwarz, we have
 $2 \sum x_i y_i \leq 2|\sum x_i y_i| \leq 2\|x\|\|y\|$. So:

$$(0 \leq) \|x+y\|^2 \leq (\|x\| + \|y\|)^2 \text{ as required.}$$

and taking $\sqrt{}$ of both sides, we preserve $\leq$ and recover the triangle inequality.

$$3. a) = \frac{1/x - 2/x^2}{3 - 2/x^2} \rightarrow \frac{0 - 0}{3 - 0} = 0 \quad (\text{divide by } x^2). \quad (0)$$

$$b) = \frac{1 + 1/x^2}{4 + 1/x} \rightarrow \frac{1 + 0}{4 + 0} = \frac{1}{4} \quad "$$

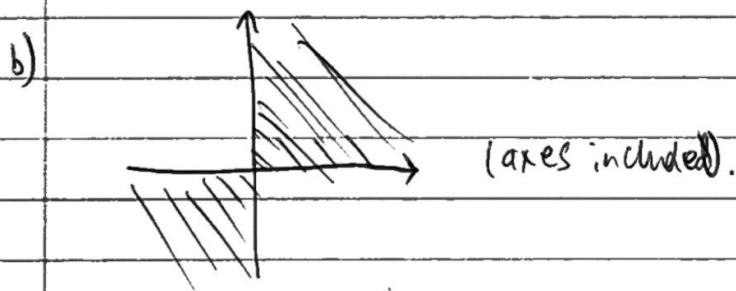
$$c) \rightarrow \frac{1}{0^{+} - 1} = -\infty$$

4. a). $A = (0, 1)$ is open, bounded, ~~compact~~, and convex. No maximum; $\sup(A) = 1$.

$B = [-\infty, 1]$ is closed and convex. $\max(B) = \sup(B) = 1$.

$C = \{x : |x| > 2\}$ is open. $\sup(C) = \infty$

$D = \mathbb{R} \setminus C$ is closed, bounded, compact, convex.
 $\max(D) = \sup(D) = 2$, no



This set is closed but unbounded; it's not convex or compact.

c) i. For any $x \in \bigcup_{i \in I} A_i$, there is an i such that $x \in A_i$, which is an open set. So there's some ϵ ball around x contained in A_i . But that ball is also contained in any union involving A_i . $\square$

ii. For any $x \in \bigcap_{i=1}^m A_i$, $\forall i: x \in A_i$, and in each case there's an open ball around x contained in A_i . So that ball must also be contained in the intersection of all of them. $\square$

Take $\epsilon = \min \{\epsilon_1, \dots, \epsilon_m\}$ as the smallest radius of all balls; defined since set is finite.

5. a) $f(x, y)$ is continuous at (x_0, y_0) iff for any $\epsilon > 0$,
 $\exists \delta > 0$ s.t. $\|(x, y)^T - (x_0, y_0)^T\| < \delta$
 $\Rightarrow \|f(x, y) - f(x_0, y_0)\| < \epsilon.$

$\Rightarrow$

Here, we want $\sqrt{x^2 + y^2} < \delta \Rightarrow |x|, |y| < \delta$

So let $\epsilon = \delta$ By triangle inequality, $|x|, |y| \leq \sqrt{x^2 + y^2}$
 $|x|, |y| \leq \sqrt{x^2 + y^2}$ since $x^2, y^2 < x^2 + y^2$.
 ~~$|x|, |y| \leq \sqrt{x^2 + y^2}$ since $x^2, y^2 < x^2 + y^2$~~

So $|x|, |y| \leq \delta$. And $|x||y| = |xy|$ for reals.

Hence, let $\delta = \sqrt{\epsilon}$. $\square$

b) We need for any $\epsilon > 0$, $\exists \delta > 0$ s.t.

~~$\|f(x) - f(x_0) + g(x) - g(x_0)\| < \epsilon$~~

$$\|x - x_0\| < \delta \Rightarrow \| \underbrace{f(x) - f(x_0) + g(x) - g(x_0)}_{=: D} \| < \epsilon.$$

By triangle inequality, $\|f(x) - f(x_0)\| + \|g(x) - g(x_0)\| < \epsilon$.

By continuity of f and g , we have some δ_f, δ_g that makes each of those terms $< \epsilon/2$ in some δ_i region around x_0 .

So we can make their sum $< \epsilon$, by choosing $\delta = \min\{\delta_f, \delta_g\}$. $\square$

c) Continuity of f means $\forall \epsilon > 0$, $\exists \delta > 0$ s.t.

$$f(B_\delta(x_0)) \subset B_\epsilon(f(x_0)) \quad \forall x_0 \in \mathbb{R}^n.$$

Consider some $w_0 \in f^{-1}(A) \subset \mathbb{R}^n$. There's some neighbourhood U around w_0 where we know $f(w_0) \in A$, which is open. So there's some region around $f(w_0)$, call it $B_\epsilon(f(w_0))$, contained in A , for an $\epsilon > 0$.

By continuity, we have a δ for this ϵ s.t. in that δ -region around w_0 , those points are mapped to the ϵ -region around $f(w_0)$.

$$\text{So } f(B_\delta(w_0)) \subset B_\epsilon(f(w_0)) \subset A.$$

Thus $B_\delta(w_0)$ is ^{contained} in the pre-image of A under f , for any arbitrary $w_0 \in f^{-1}(A)$. So $f^{-1}(A)$ is an open set. $\square$
(with $\delta > 0$).

6. Brouwer's fixed point theorem applies to functions $f: [a,b] \rightarrow [a,b]$ continuous on the closed interval $[a,b]$.

g is a polynomial, so it's continuous. In $[0,2]$, $g(x)$ starts at 0 and ends at 0, with at most one turning point. $\rightarrow$

$g''(x) = 4 - 4x$, so it attains maximum when $x = 1$ of 2. So the conditions are met.

We want to solve $x = 2x(2-x)$

$\Leftrightarrow x(3-2x) = 0$ so $x = 0$ or $3/2$ are fixed points

7. a) ~~fix~~

$$\left. \frac{\partial f}{\partial x} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \Big|_{(0,0)} \text{ if the lim exists. } \rightarrow$$

$$= \lim_{h \rightarrow 0} \frac{\frac{(h+h)^3}{(h+h)^2} - \frac{h^3}{h^2}}{h} = 1.$$

$$\text{Similarly, } \left. \frac{\partial f}{\partial y} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{0/h^2 - 0}{h} = 0$$

$$\left. \frac{\partial g}{\partial x} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{h^{1/2} \cdot 0 - 0}{h} = 0 \text{ same as above}$$

$$\left. \frac{\partial g}{\partial y} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{h^{1/2} \cdot 0 - 0}{h} = 0 \text{ also.}$$

$$3x^2(3x^2y^2) - 4x^4 \cdot 2x^4$$

b) A function is C^1 if all its partials exist and are continuous.

For f , away from $(0,0)$ its partials are

$$\frac{\partial f}{\partial x} = \frac{x^2(x^2+3y^2)}{(x^2+y^2)^2} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{-2yx^3y}{(x^2+y^2)^2}$$

But consider $\frac{\partial f}{\partial x}$ along the line $x=0$; this $\rightarrow 0$ as $y \rightarrow 0$, which $\neq 1$. So this partial isn't continuous $\Rightarrow f$ is not C^1 .

$$\text{For } g, \quad \frac{\partial g}{\partial x} = \frac{1}{2} x^{-1/2} \cdot y^{1/2}; \quad \frac{\partial g}{\partial y} = \frac{1}{2} x^{1/2} y^{-1/2}.$$

Consider along the line $y=x$. Then $\frac{\partial g}{\partial x} \rightarrow \frac{1}{2}$ as $y \rightarrow 0$, which $\neq 0$. So again, not C^1 .

c) f is differentiable at $(0,0)$ if there exists some 2×2 matrix A s.t.

$$\lim_{\|h\| \rightarrow 0} \frac{\|f(h) - f(0,0) - Ah\|}{\|h\|} = 0$$

Let $A = Df(x) = \begin{pmatrix} 1 & 0 \end{pmatrix}$. Then evaluate

$$\lim_{\|h\| \rightarrow 0} \frac{|h_1^3/h_1^2+h_2^2 - 0 - h_1|}{\sqrt{h_1^2+h_2^2}}. \quad \text{Consider along the line}$$

$$t = h_1 = h_2 \rightarrow 0^+; \quad \text{we get} \quad \frac{|1 - 1/2 t|}{\sqrt{2} t} \approx \frac{1}{2\sqrt{2}} \neq 0.$$

So condition fails (if A existed, it would be $Df(x)$); f is not differentiable.

Now For g , with $A = \begin{pmatrix} 0 & 0 \end{pmatrix}$ evaluate

$$\lim_{\|h\| \rightarrow 0} \frac{|h_1^{1/2} h_2^{1/2} - 0|}{\|h\|}. \quad \text{Again, consider the line } t = h_1 = h_2 \rightarrow 0^+.$$

We get $|t|/\sqrt{2}t = 1/\sqrt{2} \neq 0$. So $\nexists A$, not differentiable.

$$8a) D u(x) = (\partial u / \partial x_1, \dots, \partial u / \partial x_L)$$

$$= (\alpha_1 / x_1, \dots, \alpha_L / x_L)$$

$$b) D^2 u = \begin{pmatrix} \partial^2 u / \partial x_1^2 & \dots & \partial^2 u / \partial x_1 \partial x_L \\ \vdots & \ddots & \vdots \\ \partial^2 u / \partial x_L \partial x_1 & \dots & \partial^2 u / \partial x_L^2 \end{pmatrix}$$

$$= \text{diag}(-\alpha_1 \cdot x_1^{-2}, \dots, -\alpha_L \cdot x_L^{-2})$$

c) Symmetry holds immediately since diagonal.

Applying the Sylvester criterion, we can use the fact that for triangular matrices, the determinant = product of diagonal entries. So each leading principal minor is $\det(\text{some diag matrix})$ and the sequence is

$$(-1)^s \prod_{i=1}^s \alpha_i \cdot x_i^{-2}$$

which alternates in

sign, starting < 0 as required,

since $\alpha_i, x_i > 0$.

So it's -ve definite.

d) Let y^T be a nonzero $n \times 1$ vector. Then $y^T D^2 u y =$

$$\sum_{i=1}^n y_i^2 \cdot \frac{-\alpha_i}{x_i^2} < 0 \quad \text{since } y_i^2 > 0 \text{ for some } i \text{ and } > 0 \text{ for all } i, \\ \text{and } \alpha_i / x_i^2 > 0$$

So -ve definite.

$$e) u(w) \approx u(\underline{1}) + Df u(\underline{1})(w - \underline{1}) + \frac{1}{2} (w - \underline{1})^T D^2 u(\underline{1})(w - \underline{1}).$$

$$A) = 0 + \sum_{i=1}^L \alpha_i / x_i \cdot (w_i - 1) - \frac{1}{2} \sum_{i=1}^L \frac{\alpha_i}{x_i^2} \cdot (w_i - 1)^2$$

$\uparrow = 2$
 $\uparrow = 1$

$$= \sum \alpha_i w_i - 1 - \frac{1}{2} \sum \alpha_i (w_i - 1)^2$$

q. $z^2 + xz + yx^2 + y^3 = 0.$

Since $\sum \alpha_i = 1$

a) ~~Both~~ $F(x, y, z)$ is C^1 , since it's a polynomial.

$$\left. \frac{\partial F}{\partial z} \right|_{(1,0,0)} = \left. 2z + x \right|_{(1,0,0)} = 1 \neq 0.$$

So we can apply the IFT, since $F(s^*) = 0^2 + 1 \cdot 0 + 0 \cdot 1^2 + 0^3 = 0$.
 There's a C^1 function $z = g(x, y)$ defined in a neighbourhood of $(1, 0)$ satisfying:

i) $g(1, 0) = 0$

ii) $F(x, y, g(x, y)) = 0$ in that neighbourhood

iii) $\frac{\partial z}{\partial w} = - \frac{\partial F / \partial w}{\partial F / \partial z}$ for $w \in \{x, y\}$

b) ~~By IFT~~ Note $\frac{\partial F}{\partial y} = 2xy + 3y^2$; $\frac{\partial F}{\partial x} = z + 2xy$.

By (iii), $\left. \frac{\partial z(x, y)}{\partial x} \right|_{(1,0)} = - \frac{z + 2xy}{2z + x} \Big|_{(1,0)} = - \frac{0}{1} = 0$

Similarly, $\left. \frac{\partial z}{\partial y} \right|_{(1,0)} = - \frac{x^2 + 3y^2}{2z + x} \Big|_{(1,0)} = - \frac{1}{1} = -1$.

10.
$$\begin{cases} x^2 - y^2 - u^3 + v^2 + 4 = 0 \\ 2xy + y^2 - 2u^2 + 3v^4 + 9 = 0 \end{cases} \quad \underline{y}^* = (2, -1, 2, 1).$$

a) Let $\underline{x} = (x, y)$, $\underline{y} = (u, v)$. Let F be defined as above, which maps $\mathbb{R}^4 \rightarrow \mathbb{R}^2$.

$$D_{\underline{y}} F = \begin{pmatrix} \partial F_1 / \partial u & \partial F_1 / \partial v \\ \partial F_2 / \partial u & \partial F_2 / \partial v \end{pmatrix} = \begin{pmatrix} -3u^2 & 2v \\ -4u & 12v^3 \end{pmatrix}$$

$$\det(D_{\underline{y}} F|_{(2, -1)}) = -36u^2v^3 + 8uv \big|_{(2, -1)}$$

$$= -36 \times 4 \times -1 + 8 \times 2 \times -1 = -128 \neq 0$$

So $D_{\underline{y}} F$ is invertible as required for IFT.

Hence there is some function $\varphi(\underline{x})$ defined on an open ball around $(2, -1)$ s.t.

i) $\varphi(\underline{x}^*) = \underline{y}^* \quad \therefore \varphi(2, -1) = (2, 1)$

ii) $F(\underline{x}, \varphi(\underline{x})) = 0$

iii) $D_{\underline{x}} \varphi(\underline{x}) = -[D_{\underline{y}} F(\underline{x}, \varphi(\underline{x}))]^{-1} D_{\underline{x}} F(\underline{x}, \varphi(\underline{x}))$

Correct

b) $D_{\underline{x}} F|_{(2, -1)} = \begin{pmatrix} 2x & -2y \\ 2y & 2x+2y \end{pmatrix} \big|_{(2, -1)} = \begin{pmatrix} 4 & 2 \\ -2 & 2 \end{pmatrix}$

And $[D_{\underline{y}} F|_{(2, -1)}]^{-1} = \frac{1}{16} \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix} = \frac{1}{128} \begin{pmatrix} 12 & -2 \\ 8 & -12 \end{pmatrix}$

$\therefore$ by (iii), $D_{\underline{x}} \varphi(\underline{x}) = -\frac{1}{128} \begin{pmatrix} 52 & 20 \\ 56 & -8 \end{pmatrix}$

We're interested in $\partial u / \partial x$ and $\partial v / \partial x$ so take the first column.

$\partial u / \partial x = 13/32, \quad \partial v / \partial x = 7/16.$

So at $(2, -1)$ a δ change in x causes u to increase by $\approx 13/32 \delta$ and v by $\approx 7/16 \delta$.

Correct

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 $\therefore \sqrt{2}$ is not a rational number.

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$$\therefore \|x+y\|^2 = \sum_{i=1}^n 2x_i y_i + x_i^2 + y_i^2$$

$$= \|x\|^2 + \|y\|^2 + 2 \sum x_i y_i$$

$$\text{and } (\|x\| + \|y\|)^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|.$$

Norms are non-negative, so $\|x\|\|y\| \geq 0$. Since

$|2 \sum x_i y_i| \leq \|x\|\|y\|$ by Cauchy-Schwarz, we have
 $2 \sum x_i y_i \leq 2|\sum x_i y_i| \leq 2\|x\|\|y\|$. So:

$$(0 \leq) \|x+y\|^2 \leq (\|x\| + \|y\|)^2 \text{ as required.}$$

and taking $\sqrt{\quad}$ of both sides, we preserve $\leq$ and recover the triangle inequality.

3. a) $= \frac{1/x - 2/x^2}{3 - 2/x^2} \rightarrow \frac{0 - 0}{3 - 0} = 0$ (divide by x^2). (0)

b) $= \frac{1 + 1/x^2}{4 + 1/x} \rightarrow \frac{1 + 0}{4 + 0} = \frac{1}{4}$ "

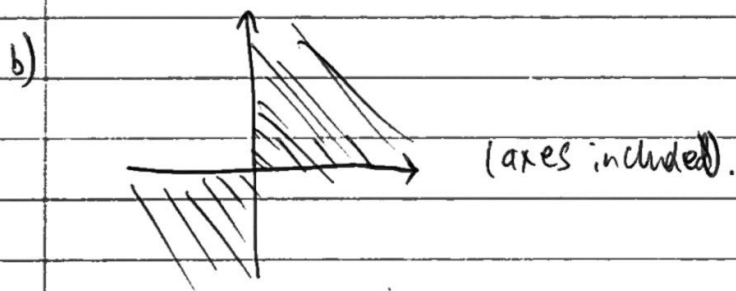
c) $\rightarrow \frac{1}{0^{+} - 1} = -\infty$

4. a). $A = (0, 1)$ is open, bounded, ~~compact~~, and convex. No maximum; $\sup(A) = 1$.

$B = [-\infty, 1]$ is closed and convex. $\max(B) = \sup(B) = 1$.

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This set is closed but unbounded; it's not convex or compact.

c) i. For any $x \in \bigcup_{i \in I} A_i$, there is an i such that $x \in A_i$, which is an open set. So there's some ϵ ball around x contained in A_i . But that ball is also contained in any union involving A_i . $\square$

ii. For any $x \in \bigcap_{i=1}^m A_i$, $\forall i: x \in A_i$, and in each case there's an open ball around x contained in A_i . So that ball must also be contained in the intersection of all of them. $\square$

Take $\epsilon = \min \{\epsilon_1, \dots, \epsilon_m\}$ as the smallest radius of all balls; defined since set is finite.

Q5a correct approach and choice of delta

5. a) $f(x, y)$ is continuous at (x_0, y_0) iff for any $\epsilon > 0$,
 $\exists \delta > 0$ s.t. $\|(x, y)^T - (x_0, y_0)^T\| < \delta$
 $\Rightarrow \|f(x, y) - f(x_0, y_0)\| < \epsilon.$

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Here, we want $\sqrt{x^2 + y^2} < \delta \Rightarrow |x|, |y| < \delta$

So let $\epsilon = \delta$ By triangle inequality, $|x|, |y| \leq \sqrt{x^2 + y^2}$
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So $|x|, |y| \leq \delta^2$. And $|x||y| = |xy|$ for reals.

Hence, let $\delta = \sqrt{\epsilon}$. $\square$

- b) We need for any $\epsilon > 0$, $\exists \delta > 0$ s.t.

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$$\|x - x_0\| < \delta \Rightarrow \| \underbrace{f(x) - f(x_0) + g(x) - g(x_0)}_{=: D} \| < \epsilon.$$

Q5b yes, correct

By triangle inequality, $\|f(x) - f(x_0)\| + \|g(x) - g(x_0)\|$

By continuity of f and g , we have some δ_f, δ_g that makes each of those $< \epsilon/2$ in some δ_i region around x_0 .

So we can make their sum $< \epsilon$, by choosing $\delta = \min\{\delta_f, \delta_g\}$. $\square$

- c) Continuity of f means $\forall \epsilon > 0$, $\exists \delta > 0$ s.t.
 ~~$f(B_\delta(x_0)) \subset B_\epsilon(f(x_0))$~~ $f(B_\delta(x_0)) \subset B_\epsilon(f(x_0)) \quad \forall x_0 \in \mathbb{R}^n$

Consider some $w_0 \in f^{-1}(A) \subset \mathbb{R}^n$. There's some neighbourhood ~~around~~ w_0 where we know $f(w_0) \in A$, which is open. So there's some region around $f(w_0)$, call it $B_\epsilon(f(w_0))$, contained in A , for an $\epsilon > 0$.

By continuity, we have a δ for this ϵ s.t. in that δ -region around w_0 , those points are mapped to the ϵ -region around $f(w_0)$.

Q5c yes, correct

$$\text{So } f(B_\delta(w_0)) \subset B_\epsilon(f(w_0)) \subset A.$$

Thus $B_\delta(w_0)$ is ^{contained} in the pre-image of A under f , for any arbitrary $w_0 \in f^{-1}(A)$. So $f^{-1}(A)$ is an open set. $\square$
(with $\delta > 0$).

6. Brouwer's fixed point theorem applies to functions $f: [a,b] \rightarrow [a,b]$ continuous on the closed interval $[a,b]$.

g is a polynomial, so it's continuous. In $[0,2]$, $g(x)$ starts at 0 and ends at 0, with at most one turning point. $\rightarrow$

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$\Leftrightarrow x(3-2x) = 0$ so $x = 0$ or $3/2$ are fixed points

7. a) ~~fix~~

$$\left. \frac{\partial f}{\partial x} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \Big|_{(0,0)} \text{ if the lim exists. } \rightarrow$$

$$= \lim_{h \rightarrow 0} \frac{\frac{(h+h)^3}{(h+h)^2} - \frac{h^3}{h^2}}{h} = 1.$$

$$\text{Similarly, } \left. \frac{\partial f}{\partial y} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{0/h^2 - 0}{h} = 0$$

$$\left. \frac{\partial g}{\partial x} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{h^{1/2} \cdot 0 - 0}{h} = 0 \text{ same as above}$$

$$\left. \frac{\partial g}{\partial y} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{h^{1/2} \cdot 0 - 0}{h} = 0 \text{ also. } \rightarrow$$

$$3x^2(1+x^2+y^2) - 4x^4 \cdot 2x^4$$

b) A function is C^1 if all its partials exist and are continuous.

For f , away from $(0,0)$ its partials are

$$\frac{\partial f}{\partial x} = \frac{x^2(x^2+3y^2)}{(x^2+y^2)^2} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{-2yx^3y}{(x^2+y^2)^2}$$

But consider $\frac{\partial f}{\partial x}$ along the line $x=0$; this $\rightarrow 0$ as $y \rightarrow 0$, which $\neq 1$. So this partial isn't continuous $\Rightarrow f$ is not C^1 .

For g , $\frac{\partial g}{\partial x} = \frac{1}{2} x^{-1/2} \cdot y^{1/2}$; $\frac{\partial g}{\partial y} = \frac{1}{2} x^{1/2} y^{-1/2}$.

Consider along the line $y=x$. Then $\frac{\partial g}{\partial x} \rightarrow \frac{1}{2}$ as $y \rightarrow 0$, which $\neq 0$. So again, not C^1 .

c) f is differentiable at $(0,0)$ if there exists some 2×2 matrix A s.t.

$$\lim_{\|h\| \rightarrow 0} \frac{\|f(h) - f(0,0) - Ah\|}{\|h\|} = 0$$

Let $A = Df(x) = \begin{pmatrix} 1 & 0 \end{pmatrix}$. Then evaluate

$$\lim_{\|h\| \rightarrow 0} \frac{|h_1^3/h_1^2+h_2^2 - 0 - h_1|}{\sqrt{h_1^2+h_2^2}}. \quad \text{Consider along the line}$$

$$t = h_1 = h_2 \rightarrow 0^+; \text{ we get } \frac{|1 - 1/2 t|}{\sqrt{2} t} \approx \frac{1}{2\sqrt{2}} \neq 0.$$

So condition fails (if A existed, it would be $Df(x)$); f is not differentiable.

Now For g , with $A = \begin{pmatrix} 0 & 0 \end{pmatrix}$ evaluate

$$\lim_{\|h\| \rightarrow 0} \frac{|h_1^{1/2} h_2^{1/2} - 0|}{\|h\|}. \quad \text{Again, consider the line } t = h_1 = h_2 \rightarrow 0^+.$$

We get $|t|/\sqrt{2}t = 1/\sqrt{2} \neq 0$. So $\nexists A$, not differentiable.

$$8a) D u(x) = (\partial u / \partial x_1, \dots, \partial u / \partial x_L)$$

$$= (\alpha_1 / x_1, \dots, \alpha_L / x_L)$$

$$b) D^2 u = \begin{pmatrix} \partial^2 u / \partial x_1^2 & \dots & \partial^2 u / \partial x_1 \partial x_L \\ \vdots & \ddots & \vdots \\ \partial^2 u / \partial x_L \partial x_1 & \dots & \partial^2 u / \partial x_L^2 \end{pmatrix}$$

$$= \text{diag}(-\alpha_1 \cdot x_1^{-2}, \dots, -\alpha_L \cdot x_L^{-2})$$

c) Symmetry holds immediately since diagonal.

Applying the Sylvester criterion, we can use the fact that for triangular matrices, the determinant = product of diagonal entries. So each leading principal minor is $\det(\text{some diag matrix})$ and the sequence is

$$(-1)^s \prod_{i=1}^s \alpha_i \cdot x_i^{-2}$$

which alternates in

sign, starting < 0 as required,

since $\alpha_i, x_i > 0$.

So it's -ve definite.

d) Let y^T be a nonzero $n \times 1$ vector. Then $y^T D^2 u y =$

$$\sum_{i=1}^n y_i^2 \cdot \frac{-\alpha_i}{x_i^2} < 0 \quad \text{since } y_i^2 > 0 \text{ for some } i \text{ and } > 0 \text{ for all } i, \\ \text{and } \alpha_i / x_i^2 > 0$$

So -ve definite.

$$e) u(w) \approx u(\underline{1}) + Df u(\underline{1})(w - \underline{1}) + \frac{1}{2} (w - \underline{1})^T D^2 u(\underline{1})(w - \underline{1}).$$

$$A) = 0 + \sum_{i=1}^L \alpha_i / x_i \cdot (w_i - 1) - \frac{1}{2} \sum_{i=1}^L \frac{\alpha_i}{x_i^2} \cdot (w_i - 1)^2$$

$\uparrow = 2$
 $\uparrow = 1$

$$= \sum \alpha_i w_i - 1 - \frac{1}{2} \sum \alpha_i (w_i - 1)^2$$

q. $z^2 + xz + yx^2 + y^3 = 0.$

Since $\sum \alpha_i = 1$

a) ~~Both~~ $F(x, y, z)$ is C^1 , since it's a polynomial.

$$\left. \frac{\partial F}{\partial z} \right|_{(1,0,0)} = \left. 2z + x \right|_{(1,0,0)} = 1 \neq 0.$$

So we can apply the IFT, since $F(s^*) = 0^2 + 1 \cdot 0 + 0 \cdot 1^2 + 0^3 = 0$.
 There's a C^1 function $z = g(x, y)$ defined in a neighbourhood of $(1, 0)$ satisfying:

i) $g(1, 0) = 0$

ii) $F(x, y, g(x, y)) = 0$ in that neighbourhood

iii) $\frac{\partial z}{\partial w} = - \frac{\partial F / \partial w}{\partial F / \partial z}$ for $w \in \{x, y\}$

b) ~~By IFT~~ Note $\frac{\partial F}{\partial y} = 2xy + 3y^2$; $\frac{\partial F}{\partial x} = z + 2xy$.

By (iii), $\left. \frac{\partial z(x, y)}{\partial x} \right|_{(1,0)} = - \frac{z + 2xy}{2z + x} \Big|_{(1,0)} = - \frac{0}{1} = 0$

Similarly, $\left. \frac{\partial z}{\partial y} \right|_{(1,0)} = - \frac{x^2 + 3y^2}{2z + x} \Big|_{(1,0)} = - \frac{1}{1} = -1$.

10.
$$\begin{cases} x^2 - y^2 - u^3 + v^2 + 4 = 0 \\ 2xy + y^2 - 2u^2 + 3v^4 + 9 = 0 \end{cases} \quad y^* = (2, -1, 2, 1).$$

a) Let $\underline{x} = (x, y)$, $\underline{y} = (u, v)$. Let F be defined as above, which maps $\mathbb{R}^4 \rightarrow \mathbb{R}^2$.

$$D_y F = \begin{pmatrix} \partial F_1 / \partial u & \partial F_1 / \partial v \\ \partial F_2 / \partial u & \partial F_2 / \partial v \end{pmatrix} = \begin{pmatrix} -3u^2 & 2v \\ -4u & 12v^3 \end{pmatrix}$$

$$\det(D_y F|_{(2, -1)}) = -36u^2v^3 + 8uv \Big|_{(2, -1)}$$

$$= -36 \times 4 \times -1 + 8 \times 2 \times -1 = -128 \neq 0$$

So $D_y F$ is invertible as required for IFT.

Hence there is some function $\varphi(\underline{x})$ defined on an open ball around $(2, -1)$ s.t.

i) $\varphi(\underline{x}^*) = y^* \quad \therefore \varphi(2, -1) = (2, 1)$

ii) $F(\underline{x}, \varphi(\underline{x})) = 0$

iii) $D_x \varphi(\underline{x}) = -[D_y F(\underline{x}, \varphi(\underline{x}))]^{-1} D_x F(\underline{x}, \varphi(\underline{x}))$

b) $D_x F|_{(2, -1)} = \begin{pmatrix} 2x & -2y \\ 2y & 2x+2y \end{pmatrix} \Big|_{(2, -1)} = \begin{pmatrix} 4 & 2 \\ -2 & 2 \end{pmatrix}$

And $[D_y F|_{(2, -1)}]^{-1} = \frac{1}{16} \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix} = \frac{1}{128} \begin{pmatrix} 12 & -2 \\ 8 & -12 \end{pmatrix}$

$\therefore$ by (iii), $D_x \varphi(\underline{x}) = -\frac{1}{128} \begin{pmatrix} 52 & 20 \\ 56 & -8 \end{pmatrix}$

We're interested in $\partial u / \partial x$ and $\partial v / \partial x$ so take the first column.

$\partial u / \partial x = 13/32, \quad \partial v / \partial x = 7/16.$

So at $(2, -1)$ a δ change in x causes u to increase by $\approx 13/32 \delta$ and v by $\approx 7/16 \delta$.

Sheet 2 - calculus

1. Suppose $p^2/q^2 = 2$, where $a^2 := p^2/q^2$ is a fraction in its simplest form.

Then $p^2 = 2q^2$ for some integers p, q . So q^2 is an int., and thus p^2 must be even. Any number whose square is even is itself even. So p is even; let $p = 2m$ for an int m .

Then $q^2 = p^2/2 = 2m^2$ is even; by the same argument q is even.

But then a was not in its simplest form. $\times$
 $\therefore \sqrt{2}$ is not a rational number.

Now consider the set $S := \{x \in \mathbb{Q} : x^2 < 2\}$ or $x < 0$ and $T := \{x \in \mathbb{Q} : x^2 > 2\}$ and $x > 0$.

There does not exist any $y \in \mathbb{Q}$ s.t.

$\forall s \in S \forall t \in T \quad s \leq y \leq t$. So $\mathbb{Q}$ is not complete,

since the only such y would be $\sqrt{2} \notin \mathbb{Q}$.

2. $\|x+y\| := \sqrt{\sum_{i=1}^n (x_i+y_i)^2}$

$$\therefore \|x+y\|^2 = \sum_{i=1}^n 2x_i y_i + x_i^2 + y_i^2$$

$$= \|x\|^2 + \|y\|^2 + 2 \sum x_i y_i$$

$$\text{and } (\|x\| + \|y\|)^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|.$$

Norms are non-negative, so $\|x\|\|y\| \geq 0$. Since

$|2 \sum x_i y_i| \leq \|x\|\|y\|$ by Cauchy-Schwarz, we have
 $2 \sum x_i y_i \leq 2|\sum x_i y_i| \leq 2\|x\|\|y\|$. So:

$$(0 \leq) \|x+y\|^2 \leq (\|x\| + \|y\|)^2 \text{ as required.}$$

and taking $\sqrt{}$ of both sides, we preserve $\leq$ and recover the triangle inequality.

3. a) $= \frac{1/x - 2/x^2}{3 - 2/x^2} \rightarrow \frac{0 - 0}{3 - 0} = 0$ (divide by x^2). (0)

b) $= \frac{1 + 1/x^2}{4 + 1/x} \rightarrow \frac{1 + 0}{4 + 0} = \frac{1}{4}$ "

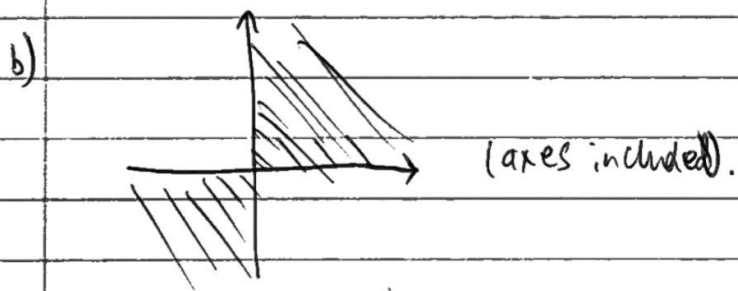
c) $\rightarrow \frac{1}{0^{+} - 1} = -\infty$

4. a). $A = (0, 1)$ is open, bounded, ~~compact~~, and convex. No maximum; $\sup(A) = 1$.

$B = [-\infty, 1]$ is closed and convex. $\max(B) = \sup(B) = 1$.

$C = \{x : |x| > 2\}$ is open. $\sup(C) = \infty$

$D = \mathbb{R} \setminus C$ is closed, bounded, compact, convex.
 $\max(D) = \sup(D) = 2$, no



This set is closed but unbounded; it's not convex or compact.

c) i. For any $x \in \bigcup_{i \in I} A_i$, there is an i such that $x \in A_i$, which is an open set. So there's some ϵ ball around x contained in A_i . But that ball is also contained in any union involving A_i . $\square$

ii. For any $x \in \bigcap_{i=1}^m A_i$, $\forall i: x \in A_i$, and in each case there's an open ball around x contained in A_i . So that ball must also be contained in the intersection of all of them. $\square$

Take $\epsilon = \min \{\epsilon_1, \dots, \epsilon_m\}$ as the smallest radius of all balls; defined since set is finite.

5. a) $f(x, y)$ is continuous at (x_0, y_0) iff for any $\epsilon > 0$,
 $\exists \delta > 0$ s.t. $\|(x, y)^T - (x_0, y_0)^T\| < \delta$
 $\Rightarrow \|f(x, y) - f(x_0, y_0)\| < \epsilon.$

$\Rightarrow$

Here, we want $\sqrt{x^2 + y^2} < \delta \Rightarrow |x|, |y| < \epsilon$

So let $\epsilon = \delta$ By triangle inequality, $|x|, |y| < \sqrt{x^2 + y^2}$
 $|x|, |y| \leq \sqrt{x^2 + y^2}$ since $x^2, y^2 < x^2 + y^2$.
 ~~$|x|, |y| \leq \sqrt{x^2 + y^2} < \delta$ since δ non-neg~~

So $|x|, |y| \leq \delta$. And $|x|, |y| = |x|, |y|$ for reals.

Hence, let $\delta = \sqrt{\epsilon}$. $\square$

b) We need for any $\epsilon > 0$, $\exists \delta > 0$ s.t.

~~$\|f(x) - f(x_0) + g(x) - g(x_0)\| < \epsilon$~~

$$\|x - x_0\| < \delta \Rightarrow \| \underbrace{f(x) - f(x_0) + g(x) - g(x_0)}_{=: D} \| < \epsilon.$$

By triangle inequality, $\|f(x) - f(x_0)\| + \|g(x) - g(x_0)\|$.

By continuity of f and g , we have some δ_f, δ_g that makes each of those $< \epsilon/2$ in some δ_i region around x_0 .

So we can make their sum $< \epsilon$, by choosing $\delta = \min\{\delta_f, \delta_g\}$. $\square$

c) Continuity of f means $\forall \epsilon > 0$, $\exists \delta > 0$ s.t.

$$f(B_\delta(x_0)) \subset B_\epsilon(f(x_0)) \quad \forall x_0 \in \mathbb{R}^n.$$

Consider some $w_0 \in f^{-1}(A) \subset \mathbb{R}^n$. There's some neighbourhood U around w_0 where we know $f(w_0) \in A$, which is open. So there's some region around $f(w_0)$, call it $B_\epsilon(f(w_0))$, contained in A , for an $\epsilon > 0$.

By continuity, we have a δ for this ϵ s.t. in that δ -region around w_0 , those points are mapped to the ϵ -region around $f(w_0)$.

$$\text{So } f(B_\delta(w_0)) \subset B_\epsilon(f(w_0)) \subset A.$$

Thus $B_\delta(w_0)$ is ^{contained} in the pre-image of A under f , for any arbitrary $w_0 \in f^{-1}(A)$. So $f^{-1}(A)$ is an open set. $\square$
(with $\delta > 0$).

6. Brouwer's fixed point theorem applies to functions $f: [a,b] \rightarrow [a,b]$ continuous on the closed interval $[a,b]$.

g is a polynomial, so it's continuous. In $[0,2]$, $g(x)$ starts at 0 and ends at 0, with at most one turning point. $\rightarrow$

$g''(x) = 4 - 4x$, so it attains maximum when $x = 1$ of 2. So the conditions are met.

We want to solve $x = 2x(2-x)$

$\Leftrightarrow x(3-2x) = 0$ so $x = 0$ or $3/2$ are fixed points

7. a) ~~fix~~

$$\left. \frac{\partial f}{\partial x} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \Big|_{(0,0)} \text{ if the lim exists. } \rightarrow$$

$$= \lim_{h \rightarrow 0} \frac{\frac{(h+h)^3}{(h+h)^2} - \frac{h^3}{h^2}}{h} = 1.$$

$$\text{Similarly, } \left. \frac{\partial f}{\partial y} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{0/h^2 - 0}{h} = 0$$

$$\left. \frac{\partial g}{\partial x} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{h^{1/2} \cdot 0 - 0}{h} = 0 \text{ same as above}$$

$$\left. \frac{\partial g}{\partial y} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{h^{1/2} \cdot 0 - 0}{h} = 0 \text{ also.}$$

$$3x^2(1+x^2+y^2) - 4x^4 \cdot 2x$$

b) A function is C^1 if all its partials exist and are continuous.

For f , away from $(0,0)$ its partials are

$$\frac{\partial f}{\partial x} = \frac{x^2(x^2+3y^2)}{(x^2+y^2)^2} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{-2yx^3y}{(x^2+y^2)^2}$$

But consider $\frac{\partial f}{\partial x}$ along the line $x=0$; this $\rightarrow 0$ as $y \rightarrow 0$, which $\neq 1$. So this partial isn't continuous $\Rightarrow f$ is not C^1 .

For g , $\frac{\partial g}{\partial x} = \frac{1}{2} x^{-1/2} \cdot y^{1/2}$; $\frac{\partial g}{\partial y} = \frac{1}{2} x^{1/2} y^{-1/2}$.

Consider along the line $y=x$. Then $\frac{\partial g}{\partial x} \rightarrow \frac{1}{2}$ as $y \rightarrow 0$, which $\neq 0$. So again, not C^1 .

c) f is differentiable at $(0,0)$ if there exists some 2×2 matrix A s.t.

$$\lim_{\|h\| \rightarrow 0} \frac{\|f(h) - f(0,0) - Ah\|}{\|h\|} = 0$$

Let $A = Df(x) = \begin{pmatrix} 1 & 0 \end{pmatrix}$. Then evaluate

$$\lim_{\|h\| \rightarrow 0} \frac{|h_1^3/h_1^2+h_2^2 - 0 - h_1|}{\sqrt{h_1^2+h_2^2}}. \quad \text{Consider along the line}$$

$$t = h_1 = h_2 \rightarrow 0^+; \text{ we get } \frac{|1 - 1/2 t|}{\sqrt{2} t} \approx \frac{1}{2\sqrt{2}} \neq 0.$$

So condition fails (if A existed, it would be $Df(x)$); f is not differentiable.

great

Now For g , with $A = \begin{pmatrix} 0 & 0 \end{pmatrix}$ evaluate

$$\lim_{\|h\| \rightarrow 0} \frac{|h_1^{1/2} h_2^{1/2} - 0|}{\|h\|}. \quad \text{Again, consider the line } t = h_1 = h_2 \rightarrow 0^+.$$

We get $|t|/\sqrt{2}t = 1/\sqrt{2} \neq 0$. So $\nexists A$, not differentiable.

$$8a) D u(x) = (\partial u / \partial x_1, \dots, \partial u / \partial x_L)$$

$$= (\alpha_1 / x_1, \dots, \alpha_L / x_L)$$

$$b) D^2 u = \begin{pmatrix} \partial^2 u / \partial x_1^2 & \dots & \partial^2 u / \partial x_1 \partial x_L \\ \vdots & \ddots & \vdots \\ \partial^2 u / \partial x_L \partial x_1 & \dots & \partial^2 u / \partial x_L^2 \end{pmatrix}$$

$$= \text{diag}(-\alpha_1 \cdot x_1^{-2}, \dots, -\alpha_L \cdot x_L^{-2})$$

c) Symmetry holds immediately since diagonal.

Applying the Sylvester criterion, we can use the fact that for triangular matrices, the determinant = product of diagonal entries. So each leading principal minor is $\det(\text{some diag matrix})$ and the sequence is

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So it's -ve definite.

d) Let y^T be a nonzero $n \times 1$ vector. Then $y^T D^2 u y =$

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a) Let $\underline{x} = (x, y)$, $\underline{y} = (u, v)$. Let F be defined as above, which maps $\mathbb{R}^4 \rightarrow \mathbb{R}^2$.

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So $D_{\underline{y}} F$ is invertible as required for IFT.

Hence there is some function $\varphi(\underline{x})$ defined on an open ball around $(2, -1)$ s.t.

i) $\varphi(\underline{x}^*) = \underline{y}^* \quad \therefore \varphi(2, -1) = (2, 1)$

ii) $F(\underline{x}, \varphi(\underline{x})) = 0$

iii) $D_{\underline{x}} \varphi(\underline{x}) = -[D_{\underline{y}} F(\underline{x}, \varphi(\underline{x}))]^{-1} D_{\underline{x}} F(\underline{x}, \varphi(\underline{x}))$

b) $D_{\underline{x}} F|_{(2, -1)} = \begin{pmatrix} 2x & -2y \\ 2y & 2x+2y \end{pmatrix} \big|_{(2, -1)} = \begin{pmatrix} 4 & 2 \\ -2 & 2 \end{pmatrix}$

And $[D_{\underline{y}} F|_{(2, -1)}]^{-1} = \frac{1}{16} \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix} = \frac{1}{128} \begin{pmatrix} 12 & -2 \\ 8 & -12 \end{pmatrix}$

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We're interested in $\partial u / \partial x$ and $\partial v / \partial x$ so take the first column.

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